DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-I
DATA MINING

Class: B. E (CSE)
Semester: V
Academic Year: 2025-2026

Max Marks: 30
Duration: 1 hr.30min
Date: 09-09-2026

		Marks	BT	CO
	PART - A Answer All Questions (Q5 x 2M = 10 M)			
1. a)	Mention any two characteristics of data mining tasks that distinguish them from traditional database queries.	2	2	5
b)	Define a data object and give one example from a banking dataset.	2	2	1
c)	What is a binary attribute? Distinguish between symmetric and asymmetric binary attributes.	2	2	1
d)	State the Euclidean distance formula for two numerical data objects.	2	1	4
e)	Why is cosine similarity particularly useful for comparing high-dimensional data such as text documents?	2	2	5
	PART - B Answer Any Two Questions (Q2 x 10M = 20 M)			
1.	a) Explain the major issues involved in data mining. Discuss challenges related to data mining methodology, user interaction, efficiency, scalability, heterogeneous data, and social or ethical concerns.	5	4	5
	b) A company wants to mine customer information containing Customer ID, Gender, Marital Status, Education Level, Age, Annual Income, and Number of Purchases. Classify each attribute and explain how the classification influences subsequent data analysis.	5	4	1
	a) Explain the similarity and dissimilarity measures for nominal, binary, ordinal, and numerical attributes. Illustrate why applying the same measure to all attribute types may produce misleading results.	5	4	5
2.	b) Consider three data objects A=(2,4,6), B=(5,4,10), and C=(2,8,6). Calculate the pairwise Euclidean distances and construct the corresponding dissimilarity matrix. Interpret the matrix.	5	4	4
	a) Explain the measures of central tendency and dispersion used for describing data. Using a suitable dataset, demonstrate the calculation and interpretation of mean, median, mode, range, variance, standard deviation, and quartiles.	5	3	1
3.	b) Two text documents are represented by the term-frequency vectors D1=(3,2,0,5) and D2=(6,4,1,10). Calculate their cosine similarity and interpret the result. Explain why cosine similarity is preferable to Euclidean distance in many text-mining applications.	5	4	5